LOTIFAR 50 MG / ML INJECTION

DESCRIPTION:
LOTIFAR 50 MG / ML INJECTION: A clear, pale straw-coloured liquid.

COMPOSITION:
LOTIFAR 50 MG/ ML INJECTION contains Netilmicin 50 mg / ml as Netilmicin Sulphate.
Contains: Sodium Metabisulphite 0.24% and Sodium Sulphite 0.08%
Preservative: Methyl Paraben 0.18%; Propyl Paraben 0.02%

PHARMACODYNAMICS:
Netilmicin is an aminoglycoside antibiotic and has a bactericidal action against many Gram-negative aerobes and against some strains of staphylococci.
Mechanism of action: Aminoglycosides are taken up into sensitive bacterial cells by an active transport process which is inhibited in anaerobic, acidic, or hyperosmolar environments. Within the cell they bind to the 30S, and to some extent to the 50S sub-units of the bacterial ribosome, inhibiting protein synthesis and generating errors in the transcription of the genetic code. The manner in which cell death is brought about in imperfectly understood, and other mechanisms may contribute, including effects on membrane permeability.
Netilmicin is an aminoglycoside antibiotic which is active against many strains of Gram-negative bacteria as well as some other bacteria including some strains of staphylococci. It exhibits synergy with beta lactams against many organisms. It is not absorbed by mouth and is usually given intramuscularly or intravenously in the treatment of severe infections, often in combination with another agent such as beta lactam. Netilmicin is not susceptible to some of the enzymes produced by organisms with acquired resistance to other aminoglycosides.

PHARMACOKINETICS:
Following intramuscular injection of netilmicin, peak plasma concentrations are achieved within half to 1 hour, and concentrations of about 7 μg per ml have been reported following doses of 2 mg per kg body-weight: similar concentrations are obtained after intravenous infusion of the same dose over 1 hour. Administration of standard, once-daily doses may produce transient peak concentrations of 20 to 30 μg per ml. The half life of netilmicin is usually 2.0 to 2.5 hours. About 80% of a dose is excreted in the urine within 24 hours.

INDICATION:
Netilmicin is effective against susceptible organisms in the following infections:
Bacteremia, septicaemia (including neonatal sepsis); serious infections of the respiratory tract; kidney and urinary tract infections; skin, soft tissue infections (including peritonitis); infected burns or wounds.
Note: Susceptibility testing should be carried out.

RECOMMENDED DOSAGE:
The recommended dosage for intramuscular and intravenous administration is identical. The patient's pre-treatment body weight should be obtained for calculation for correct dosage. Aminoglycoside dosage in obese patients should be based on an estimate of lean body mass.

The usual duration of treatment for all patients is seven to fourteen days. In complicated infections, a longer course of therapy may be necessary. Patients treated beyond the usual period should be carefully monitored for changes in renal, auditory and vestibular function. Dosage should be reduced if clinically indicated.

Patients with normal renal function:

Adults: The recommended dosage of Lotifar Injections for patients with systemic infections and normal renal function is 4-6 mg/kg/day given either as two equal doses every 12 hours or three equal doses every eight hours. For patient with life-threatening infections, dosage up to 7.5 mg/kg/day may be administered in three equal doses every eight hours. This dosage should be reduced to 6 mg/kg/day as soon as clinically indicated, usually within 48 hours. For patients with urinary tract infections without bacteraemia 4 mg/kg/day (150 mg every 12 hours) may be administered. Data on the use in the paediatric group are very limited.

The following may be used as a guide in this age group.
Children: 6.0 to 7.5 mg/kg/day (2.0 to 2.5 mg/kg administered every eight hours).
Infants and neonates over one week of age: 7.5 to 9.0 mg/kg/day (2.5 to 3.0 mg/kg administered every eight hours).
Premature or full-term neonates one week of age or less: 6 mg/kg/day (3.0 mg/kg administered every 12 hours).
Patients treated beyond the usual period should be carefully monitored for changes in renal, auditory and vestibular function. Dosage should be reduced if clinically indicated.
For intravenous administration: The intravenous administration of netilmicin is useful for treating patients with septicaemia or those in shock. It may also be the preferred route of administration for some patients with congestive heart failure, haematologic disorders, severe burns, or those with severely reduced muscle mass. Adults, a single dose of netilmicin injection may be diluted in 50 to 200 mL of sterile normal saline or in a sterile solution of glucose 5% in water; infants and children, the volume of diluent should be dependent on the patient's fluid requirements. The solution may be infused over a period of one-half to two hours. In circumstances where a prolonged infusion cannot be administered, the dose may be injected directly into a vein or IV tubing slowly over a period of 3 to 5 minutes. Netilmicin injection should not be physically premixed with other drugs, but should be administered separately in accordance with the recommended route of administration and dosage schedule. Netilmicin injection is physically compatible with the following parenteral solutions and exhibits no loss of potency at a concentration of 3 mg/mL (as the base) when stored at room temperature for one week: sterile water for injections; normal saline; 5% glucose in water; 10% glucose in water; Ringer's injections; lactated Ringer's injection.

Patients with impaired renal function: Dosage must be adjusted in patients with impaired renal function. Whenever possible, netilmicin serum concentrations should be monitored.

1. Variable frequency regimen: If serum assay determinations are not available and renal function is stable, serum creatinine and creatinine clearance values are the most readily available indicators for dosage adjustment. The following may be used as a guide only, in conjunction with close clinical and laboratory observations of the patient, and the dosage modified if deemed necessary by the treating clinician. Since the serum creatinine concentration has a high correlation with the netilmicin serum half-life, this laboratory test may be used to control netilmicin serum levels by adjusting the interval between doses. The interval between doses (in hours) may be approximated by multiplying the serum creatinine level (mg / 100 mL) by 8. For example, a patient weighing 60 kg with a serum creatinine level of 3.0 mg / 100 mL could be given 120 mg (2 mg/kg) every 24 hours. (3 x 8 hours).
2. Variable Dosage Regimen: In patients with serious systemic infections and renal impairment, the antibiotic may be administered more frequently but in reduced dosage. In such patients, netilmicin serum concentrations should be measured.

Suggested methods are:

a) After the usual initial or loading dose, a rough guide for determining reduced dosage at eight-hour intervals is to divide the normally recommended dose by the serum creatinine level. For example, after an initial dose of 120 mg (2 mg/kg), a patient weighing 60 kg with serum creatinine level of 3.0 mg / 100 mL could be given 40 mg every eight hours (120 ÷ 3).
b) If the rate of creatinine clearance is known, the maintenance dose to be administered every 8 hours may be calculated using the following formula:

$$\text{Maintenance Dose Every 8 hours} \quad \frac{\text{Observed CCr*}}{\text{Normal CCr}} \times \text{ Usual Maintenance Dose}$$

*CCr = Creatine Clearace Rate in mL/min/1.73m^2

The initial or loading dose is the same as that recommended for a patient with normal renal function. Since the status of renal function may change over the course of the infection process, netilmicin dosage may require adjustment.
Haemodialysis: In adults with renal failure undergoing haemodialysis, the amount of netilmicin removed from the blood may vary depending upon several factors, including the dialysis method used. An eighth-hour haemodialysis may reduce serum concentrations of netilmicin by approximately 50%. The recommended dose at the end of each period is 2 mg/kg. In children, a dose of 2 to 2.5 mg/kg may be administered, depending upon the severity of infection.
Concomitant therapy: Dosages recommended above for patients with normal or impaired renal function should not be reduced when netilmicin is administered concomitantly with other antibiotics.

ROUTE OF ADMINISTRATION: For I.M. and I.V. Injection

CONTRAINDICATIONS:
1. Hypersensitivity to netilmicin, or to one of the preservatives, sodium metabisulphite, sodium sulphite, sodium edetate and methyl paraben or propyl paraben.
2. A history of hypersensitivity or serious toxic reactions to other aminoglycoside antibiotics.

WARNINGS & PRECAUTIONS:
Warnings: Nephrotoxicity: Aminoglycoside antibiotics are potentially nephrotoxic. The toxicity usually takes the form of acute tubular necrosis. The risk of nephrotoxicity is increased in patients with renal impairment. Other risk factors include dehydration, concomitant treatment with potent diuretics or other nephrotoxic drugs. Patients should be well hydrated to minimise irritation of renal tubules due to the high concentration of netilmicin in the urine. Patients should be monitored for signs of renal dysfunction (BUN, serum creatinine, presence of protein or tubular cells in urine), particularly patients with renal impairment. As far as possible, serum levels should be kept within the therapeutic range of 3 - 5 mcg / mL. Prolonged serum levels above 10 mcg / mL should be avoided

Ototoxicity: Both auditory and vestibular, can occur in patients treated with netilmicin. Patients with renal impairment are at higher risk. Other patient risk factors are concomitant therapy with neurotoxic or nephrotoxic drugs and advanced age. Frequent monitoring of auditory and vestibular function is recommended. Serum levels should also be measured and serum levels kept in the range 3-5 mcg / mL. Levels above 10 mcg / mL should be avoided. Neurotoxicity: Neuromuscular blockade has been reported in animals receiving netilmicin at doses several times higher than those used clinically. However, the possibility of this phenomenon occurring in man should be considered, particularly if aminoglycosides are administered to patients receiving anaesthetic, neuromuscular blocking agents (such as succinylcholine or tubocurarine) or massive transfusions of citrate anticoagulated blood. Such blockade may be reversed by anticholinesterases or calcium salts. Patients with myasthenia gravis or with hypocalcaemia may be particularly sensitive to the neuromuscular blocking effect.

Precautions:
a) Superinfection: As with other broad-spectrum antibiotics, the use of netilmicin may result in overgrowth of non-susceptible organisms. If this occurs, appropriate therapy is indicated.
b) Monitoring of serum levels: To lessen the risk of toxicity serum levels should be monitored regularly. They should be kept in the therapeutic range 3-5 mcg /mL. Prolonged levels above 10 mcg / mL should be avoided.

This preparation contains sodium metabisulphite, a sulphite that may cause allergic-type reactions including anaphylactic symptoms and life-threatening or less severe asthmatic episodes in certain susceptible people. The overall prevalence of sulphite sensitivity is seen more frequently in asthmatic than non-asthmatic people.

INTERACTION WITH OTHER MEDICAMENTS:

Interaction with other drugs: Avoid the concurrent and / or sequential systemic use of other potentially neurotoxic and / or nephrotoxic drugs such as polymyxin B, colistin, cephaloridine, kanamycin, gentamicin, amikacin, sisomicin, tobramycin, streptomycin, paromomycin, cephalothin, viomycin, and vancomycin. Advanced age and dehydration may also increase risk of toxicity. Avoid the concurrent use of netilmicin with potent diuretics: such as ethacrynic acid or frusemide, since these diuretics by themselves may cause ototoxicity. In addition, when administered intravenously, diuretics may enhance aminoglycoside toxicity by altering the antibiotic concentration in serum and tissue.
Neurotoxic or nephrotoxic antibiotics may be absorbed from body surfaces after local irrigation or application. The potential toxic effect of such antibiotics administered in this fashion should be considered. Increased nephrotoxicty has been reported following concomitant administration of aminoglycoside antibiotics and cephalothin. The possibility of neuromuscular blockade and respiratory paralysis should be borne in mind when administering anaesthetics or neuromuscular blocking drugs in patients receiving netilmicin. A Faconi-like syndrome, with aminoaciduria and metabolic acidosis, has been reported in some adults and infants treated with netilmicin.

USE IN PREGNANCY & LACTATION:

Use in Pregnancy: entamicin and other aminoglycosides (eg. Netilmicin) are known to cross the placenta. There is evidence of selective uptake of gentamicin by the foetal kidney resulting in cellular damage (probably reversible) to immature nephrons. Eighth cranial nerve damage has also been reported following in-utero exposure to some of the aminoglycosides. Because of their chemical similarity, all aminoglycosides must be considered potentially nephrotoxic and ototoxic to the foetus. It should also be noted that therapeutic blood levels in the mother do not equate with safety for the foetus. Safety in pregnancy has not been established.

SIDE EFFECTS:

Adverse Reactions: Adverse reactions to netilmicin were similar in type and frequency to those seen with other aminoglycosides, including ototoxicity and nephrotoxicity. Local reactions included pain at side of injection, erythema and induration.
Less frequently haematoma, phlebitis and gluteal abscess were encountered.
Systemic reactions included headache, abdominal pain, vomiting, diarrhoea, palpitation, rash, drug fever, dizziness, audiometric changes, tinnitus, vertigo, disorientation, nystagmus, ringing, buzzing or sense of pressure in the ears, change in renal function such as rise in BUN and serum creatinine haematuria, eosinophilia, increase in serum bilirubin and AST and abnormalities in blood coagulation.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Overdosage: Clinical Symptoms:
a) Nephrotoxicity is characterised by an elevated BUN or serum creatinine level or a decrease in creatinine clearance.
b) Ototoxicity: Damage to vestibular function is characterised by tinnitus, dizziness or loss of hearing
c) Neurotoxicity is characterised by neuromuscular blockade with respiratory paralysis. (In some animal studies, netilmicin alone is high concentrations, caused neuromuscular blockade).

Management: In the event of toxic reactions, haemodialysis will remove netilmicin from the blood. The effectiveness of peritoneal dialysis in this regard is unknown.

STORAGE CONDITIONS:

Store below 30ºC. Do not freeze. Dark yellow solutions should not be used. Keep out of reach of children. Jauhkan daripada kanak-kanak.

SHELF LIFE:

Please refer to outer package.

PACK SIZE:

Pack in 1ml ampoule in a box of 10's, 30's, 50's and 100's.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD
Lot 2599, Jalan Seruling 59 Kawasan 3,
Taman Klang Jaya,
41200 Klang,
Selangor, MALAYSIA.